

Linux Installation and Basics*

Distributed Data Processing Environments

Lab Guide 0

This session aims at setting up a Linux environment for testing and then to learn basic shell commands, following chapters 2 to 4 of *Introdução ao Unix*.

Steps

1. Install Docker Desktop from: <https://www.docker.com/products/docker-desktop/>. On Linux, install Docker Engine instead: <https://docs.docker.com/engine/install/>. Confirm that command `docker --version` runs from CMD.
2. Create an **empty folder** to hold your work: Create folder in Explorer, click on right mouse button, and select "Open terminal here".
3. Start an Ubuntu container with that folder mounted into it:
 - Run:

```
docker run -it --name adpd-lab0 \
-v ${PWD}:/work -w /work ubuntu:24.04 bash
```
 - Files under `/work` inside the container are the files in your folder: they survive even if the container is removed.
4. Install software packages:
 - Update package lists with: `apt update`
 - Install an editor with: `apt install micro`
 - Install Python with: `apt install python3 python3-venv python3-pip`
5. Navigate the filesystem (ch. 2):
 - List the contents of `/etc` with `ls -l`.
 - Create a folder with `mkdir test`, enter it with `cd test`, and go back with `cd ..`
 - Copy `/etc/passwd` into your folder, rename the copy, and remove it with `cp`, `mv`, and `rm`.
6. Manipulate text (ch. 3), using `/etc/passwd` as input:
 - List the first 5 accounts with: `head -5 /etc/passwd`
 - Find the line describing the root account with: `grep root /etc/passwd`
 - List only the account name and shell of every user, columns 1 and 7, with:

```
cut -d: -f1,7 /etc/passwd
```
7. Compose commands (ch. 4): count how many accounts use each shell by piping the previous `cut` into `sort` and `uniq -c`:
 - Run: `cut -d: -f7 /etc/passwd | sort | uniq -c`
8. Create and use a Python virtual environment:

*No restrictions on the use of AI tools.

- Create it with: `python3 -m venv venv`
- Activate it with: `source venv/bin/activate`
- Install a package into it with: `pip install requests`
- Write a program `download.py` that uses `requests` to fetch the Bundesliga dataset¹ and save it as `matches.csv`, then run it with: `python3 download.py`
- Confirm the download with: `wc -l matches.csv`

9. At the end of the session:

- Leave the container with: `exit`
- Next time, resume the same container instead of creating a new one:
`docker start -ai adpd-lab0`
- When you no longer need it, remove the container with: `docker rm adpd-lab0`. Your files remain in the mounted folder.

Learning Outcomes Setup a locally hosted Linux environment. Apply basic shell commands to navigate the file system. Exploit text processing commands to do simple data manipulation.

¹https://huggingface.co/datasets/KrisSommer/Bundesliga_Stats_2018-2024